

Medicare Telehealth Fraud and Utilization Analytics:

An Unsupervised Anomaly Detection Framework on Post COVID HCPCS Level Provider Service Data

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*Keywords: Medicare, telehealth, fraud detection, anomaly detection, Isolation Forest, program integrity, post
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Abstract

This paper develops and validates an unsupervised anomaly detection framework for Medicare telehealth billing across the post COVID era. Using the CMS Medicare Physician and Other Practitioners by Provider and Service Public Use File for calendar year 2023 (9.66 million row level records), we construct an analytical universe of 46,900 unique providers who billed any code in a 27 code telehealth HCPCS basket. We engineer nine per provider features including Payment per Beneficiary, Services per Beneficiary, HCPCS Diversity, and Audio Only Share. A Combined Risk Score integrates a documented statistical layer (weighted absolute z scores) and an Isolation Forest machine learning layer (contamination 0.02, 100 estimators, random seed 42) at 60 percent statistical and 40 percent ML weights. Providers above the 95th percentile of the Combined Risk Score (2,346 providers, 5 percent of the universe) are flagged. The flagged set accounts for \$161.7 million in sample excess billing measured as Payment per Beneficiary above state median multiplied by beneficiary count. Bootstrap resampling at 1,000 iterations yields a 95 percent confidence interval of \$145.5M to \$179.2M. Sensitivity analysis confirms the headline finding is robust across alternative peer benchmarks (\$128.0M to \$164.0M range). Coverage adjustment using the CMS TMEDTREND Public File scales the sample to a national projection of \$557.5 million, with no recovery rate multiplier. Validation includes permutation feature importance (Audio Only Share leads at 17.5 percent); temporal holdout where 79.9 percent of providers flagged in the 2021 plus 2022 training pool reappear in the 2023 holdout with mean Payment per Beneficiary 3.35 times the population mean; OIG LEIE concordance showing 5 direct NPI matches in the flagged set, a 7.1 fold enrichment over the baseline universe rate; DOJ defendant cross reference returning zero matches as expected. All code, data file hashes, and persisted result files are publicly available. The framework emphasizes the distinction between anomaly detection and supervised fraud classification, explicitly avoiding categorical performance claims that cannot be defended without labeled ground truth.

Keywords: Medicare, telehealth, fraud detection, anomaly detection, Isolation Forest, program integrity, post COVID, HCPCS, machine learning, public use file.

JEL Classification: I11, I13, I18, H51, C45, C53.

1. Introduction

Medicare telehealth utilization expanded substantially during the COVID 19 public health emergency. The CMS Medicare Telehealth Trends Public File documents that in 2020 the share of Medicare Part B beneficiaries who used at least one telehealth service peaked at 47.91 percent, representing 14.83 million beneficiaries. By 2023 the share had normalized to 24.99 percent, still representing 6.97 million beneficiaries.

The Government Accountability Office has designated Medicare program integrity as a high risk area since 1990. The Office of Inspector General reported \$31.7 billion in Medicare improper payments in fiscal year 2024. The Department of Justice June 2025 National Health Care Fraud Takedown charged 324 defendants in schemes totaling \$14.6 billion in alleged fraud. HHS-OIG OEI-02-20-00720 documents the major risk patterns from pandemic era expansion: audio only billing concentration, single HCPCS dominance, geographic dispersion of patients, and rapid post emergency growth in providers with limited prior Medicare history.

This paper contributes an unsupervised anomaly detection framework that addresses three documented limitations of prior analytics work in this area. First, prior analytical artifacts in this domain have typically operated on the general Medicare physician summary file rather than the HCPCS level Provider and Service file, making it impossible to apply a true telehealth filter at the row level. Second, prior work has frequently claimed supervised classification metrics (precision, recall, F1, AUC) on unsupervised models without labeled ground truth, which is methodologically indefensible. Third, prior work has presented hardcoded performance numbers, fabricated feature importance tables, and decorative validation claims that do not trace back to executed code.

The framework presented here addresses all three. The analytical universe is HCPCS filtered at the row level. The validation suite uses only procedures appropriate to unsupervised methodology. Every numerical claim in the paper traces to a specific notebook cell and a persisted result file, with SHA 256 hashes of all source files recorded in a reproducibility envelope.

2. Methods

2.1 Data Sources

The primary analytical input is the CMS Medicare Physician and Other Practitioners by Provider and Service Public Use File for calendar years 2019 through 2023, containing 48,891,840 rows. Secondary sources: OIG LEIE Database (81,914 excluded entities), CMS TMEDTREND Public File, Medicare Provider Taxonomy Crosswalk. SHA 256 hashes of all source files are recorded in RUN_INFO.json.

2.2 Telehealth HCPCS Code Basket

The analytical universe is defined by 27 HCPCS codes covering five service categories. Online digital evaluation and management: 99421, 99422, 99423. Telephone evaluation and management by physician: 99441, 99442, 99443. Remote physiologic monitoring under CPT codes: 99453, 99454, 99457, 99458, 99091. Implantable cardiovascular monitor remote interrogation: G2066. Brief virtual check ins, originating site facility fees, and qualified nonphysician communication technology services: G2010,

G2012, G2061 through G2063, G2250 through G2252, Q3014, 98966 through 98968, 98970 through 98972. Filter rules: HCPCS_Cd in basket AND Tot_Benes greater than or equal to 11 AND Avg_Mdcr_Pymt_Amt greater than zero. The 2023 filtered universe contains 73,266 rows and 46,900 unique NPIs.

2.3 Feature Engineering

Filtered rows are aggregated to the provider level. Nine per provider features are computed: Telehealth_Services_Total, Telehealth_Benes_Max, Telehealth_Payment_Total, Telehealth_HCPCS_Diversity, Services_Per_Bene, Payment_Per_Bene, Avg_Payment_Per_Service, Audio_Only_Share, and Log_Services. NaN and infinity values are replaced with feature medians.

2.4 Combined Risk Score

The Statistical Risk Layer applies documented per feature weights summing to 1.0: Payment_Per_Bene 0.30, Services_Per_Bene 0.25, Avg_Payment_Per_Service 0.15, Audio_Only_Share 0.10, Telehealth_HCPCS_Diversity 0.10, Log_Services 0.10. Each feature contributes its absolute z score against the national mean and standard deviation. The Machine Learning Layer fits an Isolation Forest (Liu, Ting, and Zhou, 2008) with contamination 0.02, n_estimators 100, random_state 42 on a standardized eight feature matrix using StandardScaler. The Combined Risk Score is the min max normalized weighted sum at 60 percent statistical and 40 percent ML, rescaled to 0 to 100.

2.5 Peer Benchmark and Excess Billing

For each flagged provider, excess billing is computed as the maximum of zero and (Payment_Per_Bene minus State_Median_Payment_Per_Bene) multiplied by Telehealth_Benes_Max. State medians are computed across the full 2023 universe.

Sensitivity analysis confirms the headline excess finding is robust across alternative peer benchmarks. National median yields \$164.0M, state median (chosen) \$161.7M, specialty median \$138.1M, and specialty-state median \$128.0M. State median was chosen to capture meaningful geographic variation in Medicare reimbursement while remaining computationally tractable and interpretable.

2.6 National Projection

The national projection multiplier is derived from the CMS TMEDTREND Public File. National telehealth beneficiaries in 2023: 6,971,495. Universe Telehealth_Benes_Max sum: 2,022,543. Coverage multiplier: 3.447. Projected national excess: \$557.5 million. No recovery rate multiplier is applied.

2.7 Validation Procedures

Bootstrap resampling: 1,000 iterations with replacement. Permutation feature importance: 5 repeats per feature, drop in Spearman rank correlation as importance metric. Temporal holdout: train on 2021+2022 concatenated universe, hold out 2023, measure NPI persistence and Payment_Per_Bene lift. OIG LEIE concordance: direct NPI match plus fuzzy name match at SequenceMatcher 0.85 threshold. DOJ defendant cross reference: fuzzy match against named defendants from DOJ press releases and HHS-OIG

reports. Fraud typology alignment: five telehealth specific typologies. Threshold sensitivity: percentile cutoff iterated 90 through 99.9.

3. Results

3.1 Headline Findings

The 2023 analytical universe contains 46,900 unique Medicare providers across 57 states and 83 distinct provider types. Total telehealth services billed: 9,118,975. Total Medicare telehealth payments: \$396,167,173. Flagged providers (top 5 percent): 2,346. Sample excess billing across flagged set: \$161,742,187. Bootstrap 95 percent CI: \$145,467,900 to \$179,224,021.

3.2 Risk Tier Distribution

Tier counts: Low 37,519 (80.0 percent), Medium 7,035 (15.0 percent), High 1,876 (4.0 percent), Critical 423 (0.9 percent), Extreme 47 (0.1 percent). The Extreme tier represents the 99.9th percentile and above.

3.3 Geographic Distribution

Top five states by flagged provider excess billing: CA (\$35.3M, 439 providers), NY (\$32.2M, 265 providers), TX (\$15.6M, 300 providers), FL (\$13.3M, 192 providers), NJ (\$6.9M, 133 providers). The geographic concentration is consistent with the historical Medicare fraud hotspot states identified in HHS-OIG audit reports.

3.4 Top Outliers

The top outlier by sample excess is Cardionet, Llc in NY (Independent Diagnostic Testing Facility (IDTF)), with 78,313 telehealth services to 11,032 beneficiaries and \$3.95M sample excess. The top 10 flagged providers by excess billing are dominated by Independent Diagnostic Testing Facilities (IDTFs) specializing in remote interrogation of implantable cardiovascular monitor systems under HCPCS code G2066. This service category is a legitimate Medicare benefit. The high sample excess reflects concentrated billing in a single specialty code at volumes substantially above the broader 2023 telehealth provider population. The flagging methodology surfaces providers whose billing patterns warrant investigative review of medical necessity, beneficiary outreach practices, and pricing relative to peer remote monitoring operations; it does not imply fraud.

3.5 Temporal Holdout Validation

The 2021 plus 2022 training pool contains 99,156 unique NPIs. The pipeline flags 4,958 NPIs at the top 5 percent cutoff. Of these, 3,959 (79.9 percent) reappear in the 2023 holdout. Their mean Payment per Beneficiary in 2023 is \$462.68 versus population mean \$138.19. Mean lift ratio 3.35. Median lift ratio 4.97. The high holdout coverage and substantial lift indicate flagged providers exhibit persistent anomalous billing patterns rather than random year over year noise.

3.6 Permutation Feature Importance

Measured contributions to the Isolation Forest anomaly score, sorted: Audio_Only_Share (17.5 percent); Services_Per_Bene (16.1 percent); Telehealth_Services_Total (15.1 percent);

Telehealth_HCPCS_Diversity (13.1 percent); Payment_Per_Bene (10.0 percent); Avg_Payment_Per_Service (9.8 percent); Telehealth_Benes_Max (9.3 percent); Log_Services (9.1 percent). Audio Only Share is the top feature, consistent with HHS-OIG documented risk patterns identifying audio only telephone billing as a major post pandemic risk signal.

3.7 OIG LEIE Concordance and DOJ Cross Reference

Direct NPI exact match between the 2,346 flagged providers and the OIG LEIE database yields 5 matches. The flagged set rate of LEIE matches is 0.21 percent versus the universe baseline rate of 0.03 percent, a 7.1 fold enrichment. The enrichment indicates the Combined Risk Score is identifying providers who share characteristics with previously excluded entities. Fuzzy name matching of the top 100 flagged providers at 0.85 SequenceMatcher threshold yields 8 candidate matches, most being common name coincidences. DOJ defendant cross reference returns zero matches against the top 500 flagged providers, as expected.

3.8 Fraud Typology Pattern Alignment

Among the 2,346 flagged providers, the five telehealth specific typologies match as follows. Extreme Payment per Beneficiary: 488 (20.8 percent). Audio Only Heavy: 200 (8.5 percent). Single HCPCS Dominance: 148. Rural High Volume: 1. Pandemic Emergence: 1295 (55.2 percent), the dominant typology. 475 providers match two or more typologies.

3.9 Bootstrap Confidence Intervals

Bootstrap resampling at 1,000 iterations on the 46,900 provider universe yields a 95 percent CI for flagged provider count of [2345, 2347] and for sample excess billing of [\$145.5M, \$179.2M].

3.10 Threshold Sensitivity

At the 95th percentile cutoff: 2,346 providers flagged, \$161.7M sample excess. At the 99th percentile: 469 providers, \$71.1M excess. At the 99.9th: 47 providers, \$11.0M excess. The monotonic profile across cutoffs indicates stable analytical conclusions.

4. Discussion

4.1 Interpretation

The framework identifies a substantial population of Medicare telehealth providers whose billing patterns deviate from peer benchmarks. The flagged set is concentrated in California, New York, Texas, and Florida, which are the historical Medicare fraud hotspot states identified in OIG audit reports. The dominant fraud typology is Pandemic Emergence, with 55 percent of flagged providers showing no 2019 telehealth presence but high 2023 telehealth volume, consistent with the HHS-OIG OEI-02-20-00720 documented risk pattern.

4.2 The Top Outlier Case

The top outlier by sample excess is Cardionet, Llc (NY), an Independent Diagnostic Testing Facility specializing in remote interrogation of implantable cardiovascular monitor systems. The provider billed

78,313 services to 11,032 beneficiaries in 2023 under HCPCS code G2066, generating \$5,042,662 in Medicare payments. The provider's Payment per Beneficiary of \$457.09 substantially exceeds the New York telehealth state median. This billing pattern is characteristic of large scale remote cardiac monitoring operations under HCPCS code G2066, which is the dedicated code for interrogation of implantable cardiovascular monitor systems. G2066 became a significant Medicare payment category after CMS added it as a permanent reimbursement category. The framework flags such providers for investigative review of billing volume, pricing relative to peer remote monitoring operations, and medical necessity documentation, without making any determination of whether the billing reflects appropriate service delivery.

4.3 Implications for Federal Practice

The framework demonstrates that publicly available CMS data is sufficient to surface high anomaly providers at scale without requiring access to claim level personally identifying information. The methodology generalizes to other Medicare service categories with adapted feature engineering and domain specific fraud typologies.

4.4 Comparison to Existing Approaches

Existing federal program integrity analytics rely heavily on retrospective review and direct OIG exclusion matching. Both approaches detect fraud after it has occurred. The framework documented here surfaces anomalous billing patterns from current billing data, with the explicit framing that flags warrant investigative review rather than enforcement.

5. Limitations

- 1. Unsupervised nature.** No labeled ground truth of fraud exists. Conventional supervised metrics cannot be computed.
- 2. Beneficiary count proxy.** Telehealth_Benes_Max uses the maximum across HCPCS rows for each provider. The public use file does not enable exact beneficiary deduplication.
- 3. Peer benchmark choice.** State median was chosen as the peer benchmark for primary results. Sensitivity analysis confirms the headline excess finding ranges from \$128.0M (specialty-state median) to \$164.0M (national median) across reasonable benchmarks. The specialty-state median is more rigorous for service-category specific assessments but produces a finding in the same order of magnitude.
- 4. IDTF business model concentration.** The top outliers are Independent Diagnostic Testing Facilities specializing in remote cardiac monitoring under HCPCS code G2066, a legitimate Medicare benefit category. The flagging reflects high billing volume in a single specialty code and does not imply fraud.
- 5. Fee for service scope.** Medicare Advantage telehealth is not captured in the public use file.
- 6. Time window.** The analytical universe is calendar year 2023.
- 7. OIG LEIE structural reality.** Excluded providers are barred from billing Medicare. Five direct matches likely reflect exclusion date timing or pending reinstatement.

8. Projection methodology assumption. The coverage adjustment assumes flagged provider billing intensity is representative of the broader telehealth population.

9. Model choice. The machine learning layer uses Isolation Forest only. Multi detector ensembles were not evaluated.

6. Conclusions

This paper develops and validates an unsupervised anomaly detection framework for Medicare telehealth billing across the post COVID era. Using publicly available CMS data and documented methodology, the framework identifies 2,346 providers whose 2023 telehealth billing patterns deviate substantially from state peer benchmarks, accounting for \$161.7 million in sample excess billing (95 percent CI [\$145.5M, \$179.2M]; sensitivity range \$128.0M to \$164.0M across benchmark choices) and a coverage adjusted national projection of \$557.5 million. The validation suite uses procedures appropriate to unsupervised methodology, with temporal holdout demonstrating 79.9 percent persistence of flagged providers across years and OIG LEIE enrichment of 7.1 fold in the flagged set. The framework explicitly avoids categorical performance claims that cannot be defended without labeled ground truth. All code, source file hashes, parameters, and persisted result files are publicly available, supporting full independent reproducibility.

7. Data and Code Availability

All source data is publicly available from data.cms.gov and oig.hhs.gov under FOIA eligible public use file licenses. The analytical notebook, persisted result files, figures, dashboard, claims traceability table, and reproducibility envelope are available in the project repository.

8. Conflicts of Interest

The author declares no financial conflicts of interest. The work was performed independently on the author's personal time using publicly available CMS data.

9. References

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[9] U.S. Department of Justice. June 2025 National Health Care Fraud Takedown. \$14.6 billion, 324 defendants. justice.gov/opa/pr

[10] U.S. Government Accountability Office. GAO-23-106203, High Risk Series.

[11] Centers for Medicare and Medicaid Services. FY 2024 Improper Payments Fact Sheet.

[12] American Medical Association CPT Editorial Panel. HCPCS code G2066: Interrogation device evaluation(s), (remote) up to 30 days; implantable cardiovascular monitor system.

Appendix A: Reproducibility Envelope Excerpt

Random seed: 42. Isolation Forest contamination: 0.02. n_estimators: 100. Bootstrap iterations: 1000. Permutation repeats: 5. Fuzzy similarity threshold: 0.85. Combined score weights: statistical 0.6, ML 0.4.

Software: Python 3.10.12, numpy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, scipy 1.15.3.

Appendix B: Peer Benchmark Sensitivity

Peer benchmark choice affects the sample excess estimate. The chosen state-median benchmark yields \$161.7M. Alternative benchmarks: national median \$164.0M (slightly higher because national median is lower than some state medians); specialty median \$138.1M (lower because specialty-specific medians better capture within-category variation); specialty-state median \$128.0M (lowest because it controls for both geographic and specialty variation). The chosen state-median benchmark sits between the bounds and reflects the most interpretable comparison for cross-state Medicare program integrity assessments.

End of preprint manuscript.